Q1 Report: Multi-Stage Cepstral Feature Extraction and Phoneme Boundary Detection

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1 Objective

This report analyzes a manual speech feature extraction and boundary detection pipeline built for four required modules:

1. Manual MFCC/Cepstrum extraction (pre-emphasis, windowing, FFT, Mel filterbank, log compression, DCT).
2. Spectral leakage and SNR comparison across rectangular, Hamming, and Hanning windows.
3. Cepstrum-based voiced/unvoiced (V/UV) boundary detection.
4. Mapping manual boundaries to Hugging Face token-level alignments with RMSE evaluation.

2 Data and Experimental Setup

Audio files were taken from the provided manifest in `data/audio_manifest.csv` (three WAV files at 16 kHz target sampling rate). The reported outputs correspond to one representative utterance of length approximately 4.0 s, producing 398 analysis frames with 25 ms frame size and 10 ms hop.

Core hyperparameters

- Pre-emphasis factor: $\alpha = 0.97$.
- Frame length/hop: 25 ms/10 ms.
- MFCC FFT size: $N_{\text{FFT}} = 512$, Mel bins: 26, cepstral coefficients: 13.
- Leakage/SNR FFT size: $N_{\text{FFT}} = 1024$.
- Cepstral V/UV bands: low quefrency $1 \ldots 0.002 f_s$, high quefrency $0.003 f_s \ldots 0.015 f_s$.
- V/UV score threshold: 55th percentile of smoothed high/low ratio.
- Alignment model: `facebook/wav2vec2-base-960h` (CTC decoding).

3 Method Summary

3.1 Manual MFCC/Cepstrum Engine

Given signal $x[n]$, pre-emphasis is

$$y[n] = x[n] - \alpha x[n-1], \quad \alpha = 0.97. \tag{1}$$

Each frame is windowed (default Hamming), transformed with real FFT, and converted to power spectrum:

$$P(k) = \frac{|X(k)|^2}{N_{\text{FFT}}}. \tag{2}$$

Triangular Mel filters produce energies E_m, then

$$L_m = \log(\max(E_m, \epsilon)), \qquad c_r = \text{DCT}_{\text{II}}(L_m),\ r = 0, \ldots, 12. \tag{3}$$

In parallel, a real cepstrum is obtained via inverse FFT of log-magnitude spectrum for low/high quefrency analysis.

3.2 Spectral Leakage and SNR Tool

For a representative high-energy frame and each window $w[n]$, spectral power is partitioned into main-lobe and out-of-lobe components:

$$\text{Leakage ratio} = \frac{P_{\text{leak}}}{P_{\text{total}}}, \tag{4}$$

$$\text{SNR}_{\text{win}} = 10 \log_{10} \left(\frac{P_{\text{main}}}{P_{\text{leak}}} \right). \tag{5}$$

Main-lobe bounds are found around the spectral peak using a $-20\,\text{dB}$ threshold relative to the peak.

3.3 Voiced/Unvoiced Boundary Detection

From cepstrum magnitude per frame,

$$s_t = \frac{\overline{C}_{\text{high}}(t)}{\overline{C}_{\text{low}}(t) + \epsilon}, \tag{6}$$

then a moving average smooths s_t. Frames with s_t above the 55th percentile are marked voiced; short flips ($< 30\,\text{ms}$) are removed.

3.4 Phonetic Mapping and RMSE

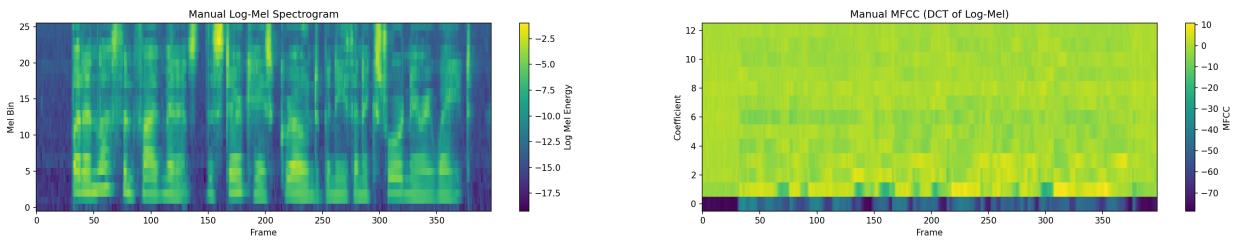
CTC argmax token runs from Wav2Vec2 are converted to token-time segments. Manual V/UV segments are mapped to dominant overlapping model token. Boundary error is measured as nearest-neighbor RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (b_i - \hat{b}_{\text{nearest}})^2}. \tag{7}$$

4 Results and Analysis

4.1 MFCC/Cepstrum Outputs

Figure 1 shows the manual log-Mel and MFCC outputs. The 0th cepstral coefficient dominates dynamic range (energy term), while higher coefficients are lower-amplitude and smoother, which matches expected MFCC behavior.



(a) Manual log-Mel spectrogram

(b) Manual MFCC matrix

Figure 1: Outputs from the handcrafted MFCC engine.

Cepstral region trends (Figure 2) show low-quefrency envelope energy consistently above high-quefrency periodicity, with local high-quefrency rises at voiced regions.

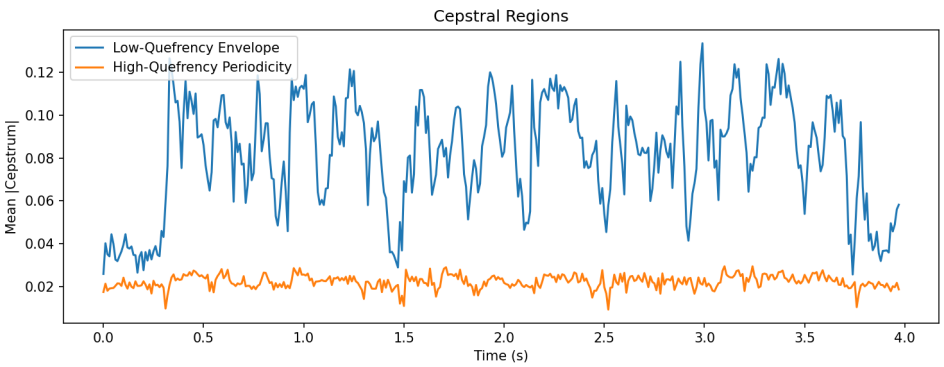


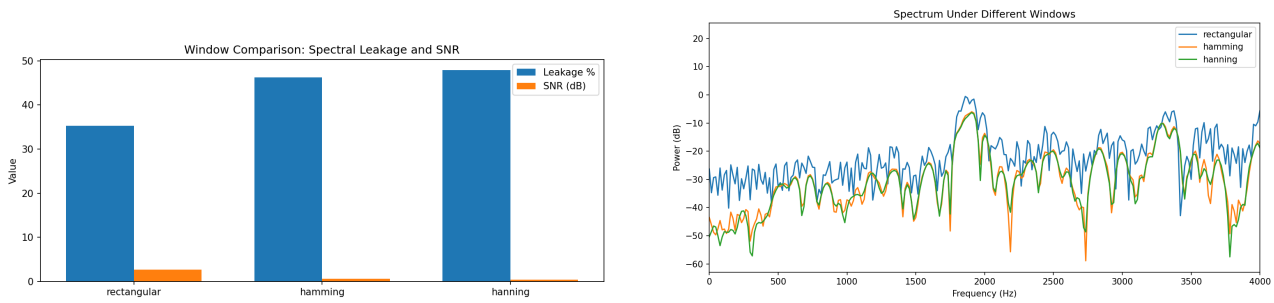
Figure 2: Low- and high-quefrency cepstral trajectories over time.

4.2 Windowing: Spectral Leakage and SNR

Measured values from `leakage_snr_table.csv` are in Table 1.

Table 1: Spectral leakage and SNR for three windows.

Window	Leakage (%)	SNR (dB)
Rectangular	35.232	2.644
Hamming	46.243	0.654
Hanning	47.867	0.371



(a) Bar comparison

(b) Power spectra

Figure 3: Leakage/SNR outputs for rectangular, Hamming, and Hanning windows.

Interpretation. In this run, rectangular appears best (lowest leakage, highest SNR), which is opposite to textbook sidelobe behavior. The most likely reasons are:

- single-frame evaluation on real speech (not a pure tone),
- main-lobe definition via a fixed $-20\,\mathrm{dB}$ threshold,
- wider main lobes for tapered windows changing power partitioning under this specific metric.

Therefore, these values are valid for the implemented metric, but should not be over-generalized as universal window ranking.

4.3 Voiced/Unvoiced Segmentation

The algorithm generated 35 contiguous segments: 18 voiced and 17 unvoiced. Total voiced duration is approximately 1.89 s, unvoiced 2.43 s, over 4.32 s analyzed time.

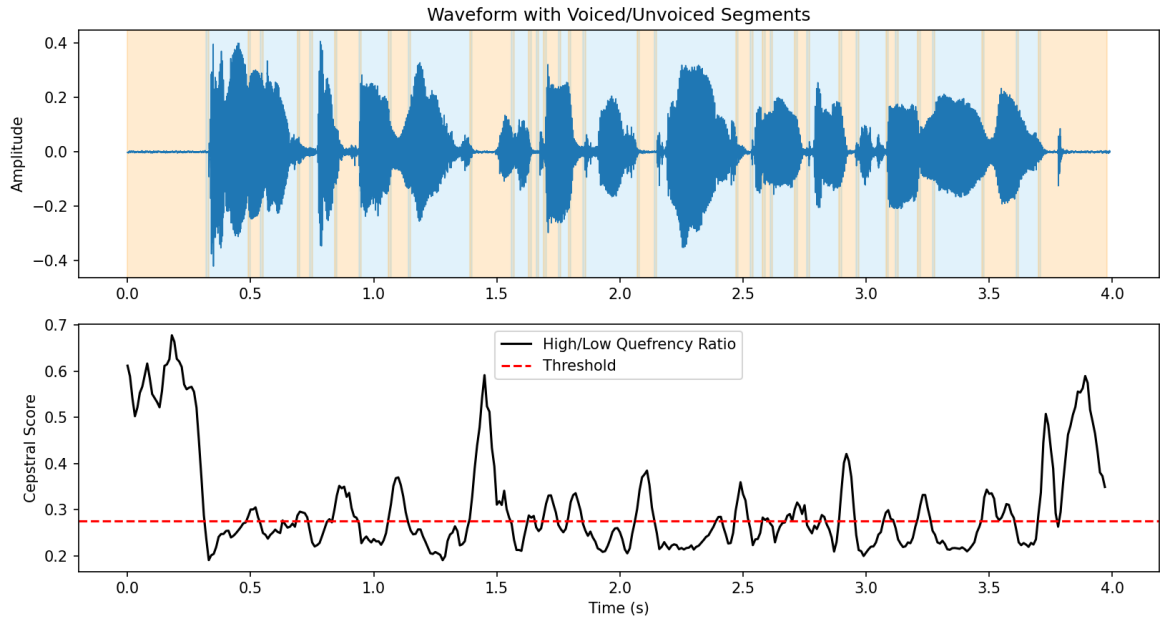


Figure 4: Waveform with V/UV regions and cepstral decision score.

Interpretation. The score tracks phonetic activity and captures many short alternations. However, early near-silence is partially labeled voiced, indicating sensitivity to threshold selection and local normalization. A practical improvement is to combine cepstral ratio with short-time energy or zero-crossing veto for silence handling.

4.4 Phonetic Mapping and Boundary RMSE

Manual segments were aligned against CTC token segments from Wav2Vec2. The output RMSE is:

$$\text{RMSE} = 0.02077\,\text{s} = 20.77\,\text{ms}. \tag{8}$$

Table 2: Boundary alignment error (from `rmse_metrics.csv`).

Metric	Value
RMSE (s)	0.02077
RMSE (ms)	20.77

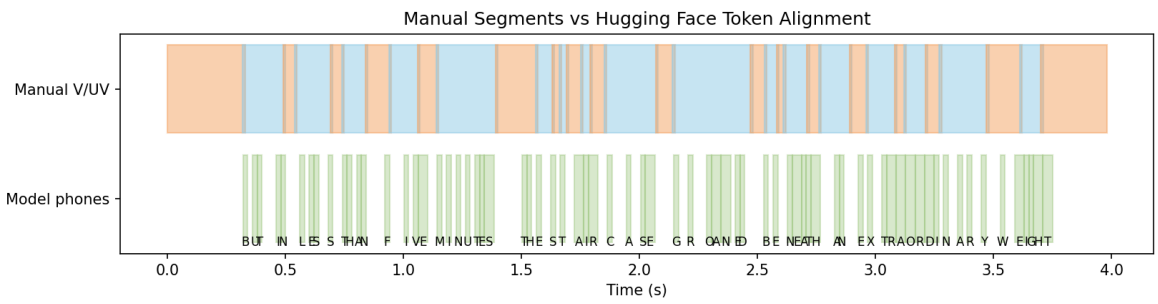


Figure 5: Manual V/UV regions (top) vs model token alignment (bottom).

Interpretation. An RMSE near 21 ms is within roughly two analysis hops (10 ms each), indicating reasonable temporal agreement for a lightweight manual method.

5 Conclusion

The key quantitative outcomes are:

- MFCC engine outputs are consistent with expected cepstral structure.
- Window metric values: rectangular (35.23%, 2.64 dB), Hamming (46.24%, 0.65 dB), Hanning (47.87%, 0.37 dB) under the current definition.
- V/UV segmentation produced 35 segments across 4.32 s.
- Manual-vs-model boundary RMSE: 20.77 ms.